# Gradating Arrow's Axioms

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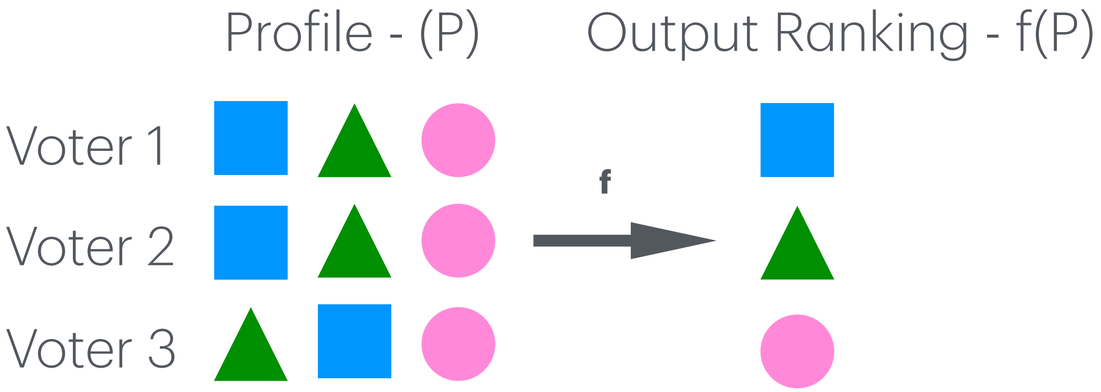
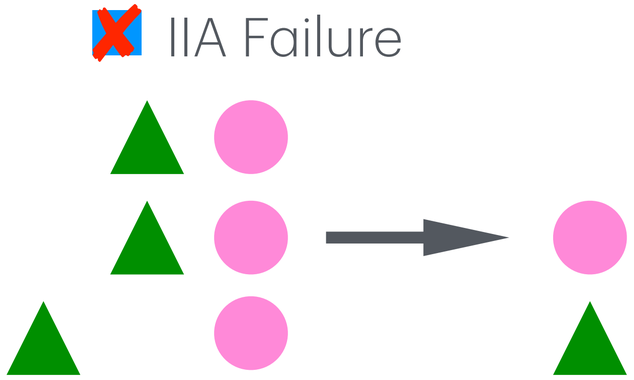
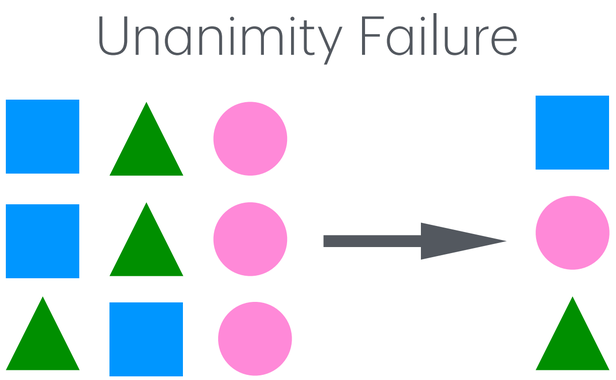
# Arrow's Theorem

- No ranking-based voting rule with 3 or more alternatives can simultaneously satisfy
  - Independence of Irrelevant Alternatives (IIA),
  - Unanimity (U),
  - Non-Dictatorship



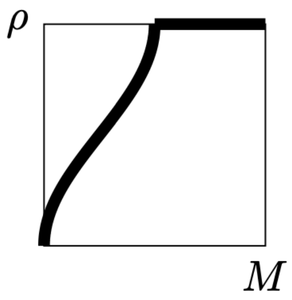



# Definitions of axiom defect

$$\rho_{\mathrm{IIA}}(f,P) := 1 - \frac{\sum_{C\in\mathcal{C}} d_{\mathrm{swap}}(f(P^C), f(P)^C)}{m\binom{m-1}{2}}.$$

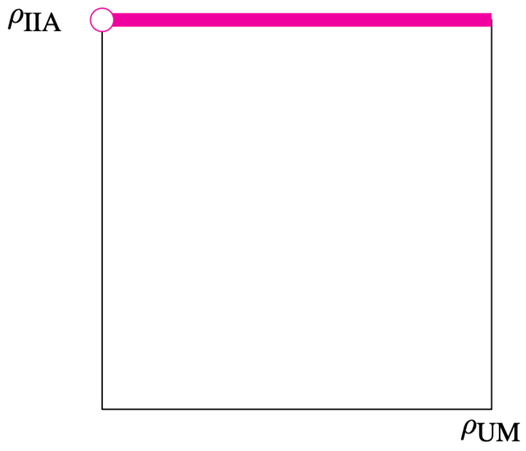
$$M(f,P) := \min_{A \succ_{f(P)} B} \frac{\#\{i \in \mathcal{V} : A \succ_i B\}}{n}.$$

$$\rho_{\mathrm{UM}}(f,P) := \begin{cases} \frac{2}{\pi}\arcsin\sqrt{2M}, & M(f,P) < 1/2 \\ 1, & M(f,P) \geq 1/2, \end{cases}$$



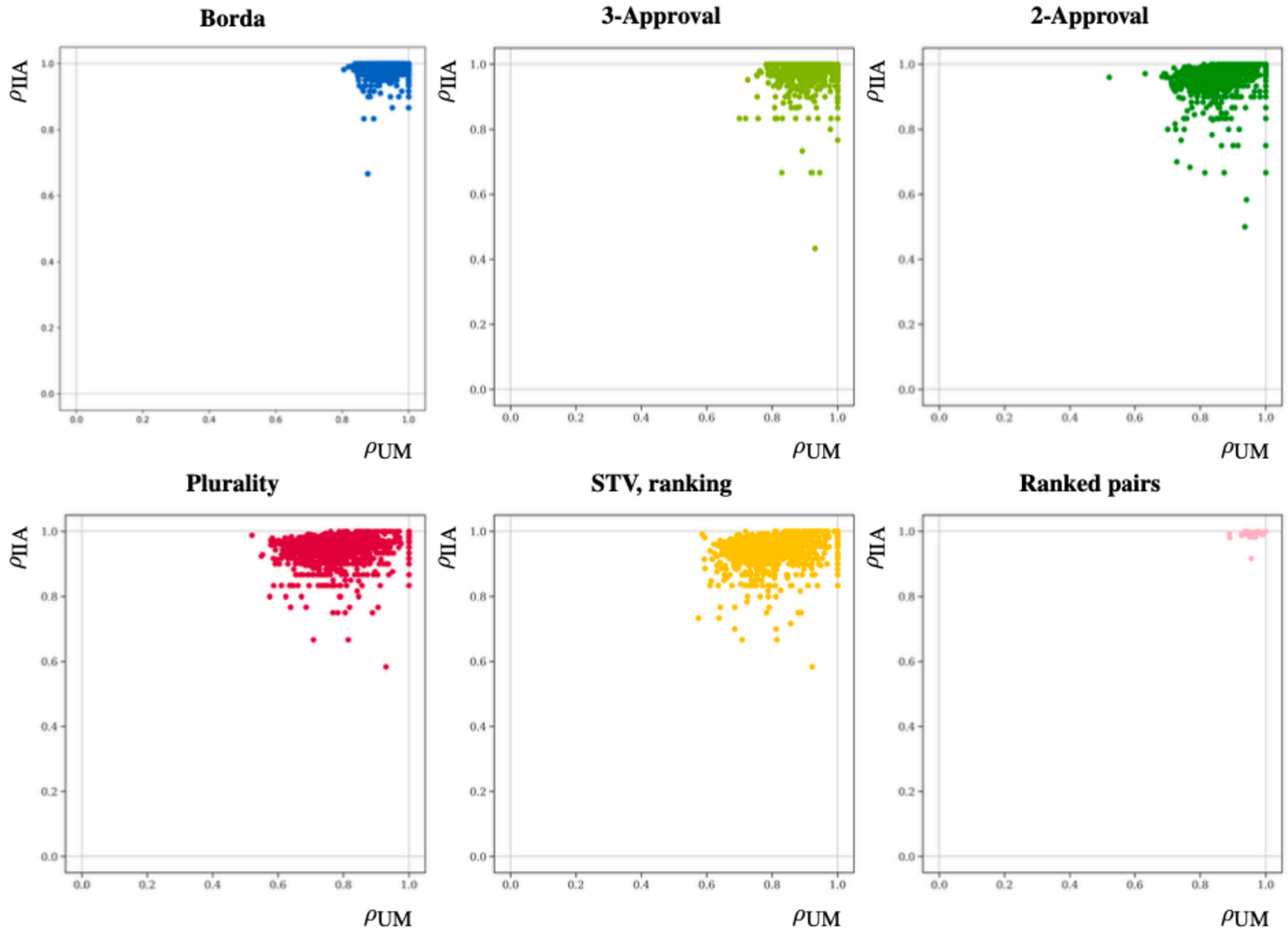

**Visual Arrow's Theorem**:

A ranking rule that always hits the **pink** strip ($\rho_{\mathrm{UM}} > 0,\ \rho_{\mathrm{IIA}} = 1$) must be **Dictatorship**



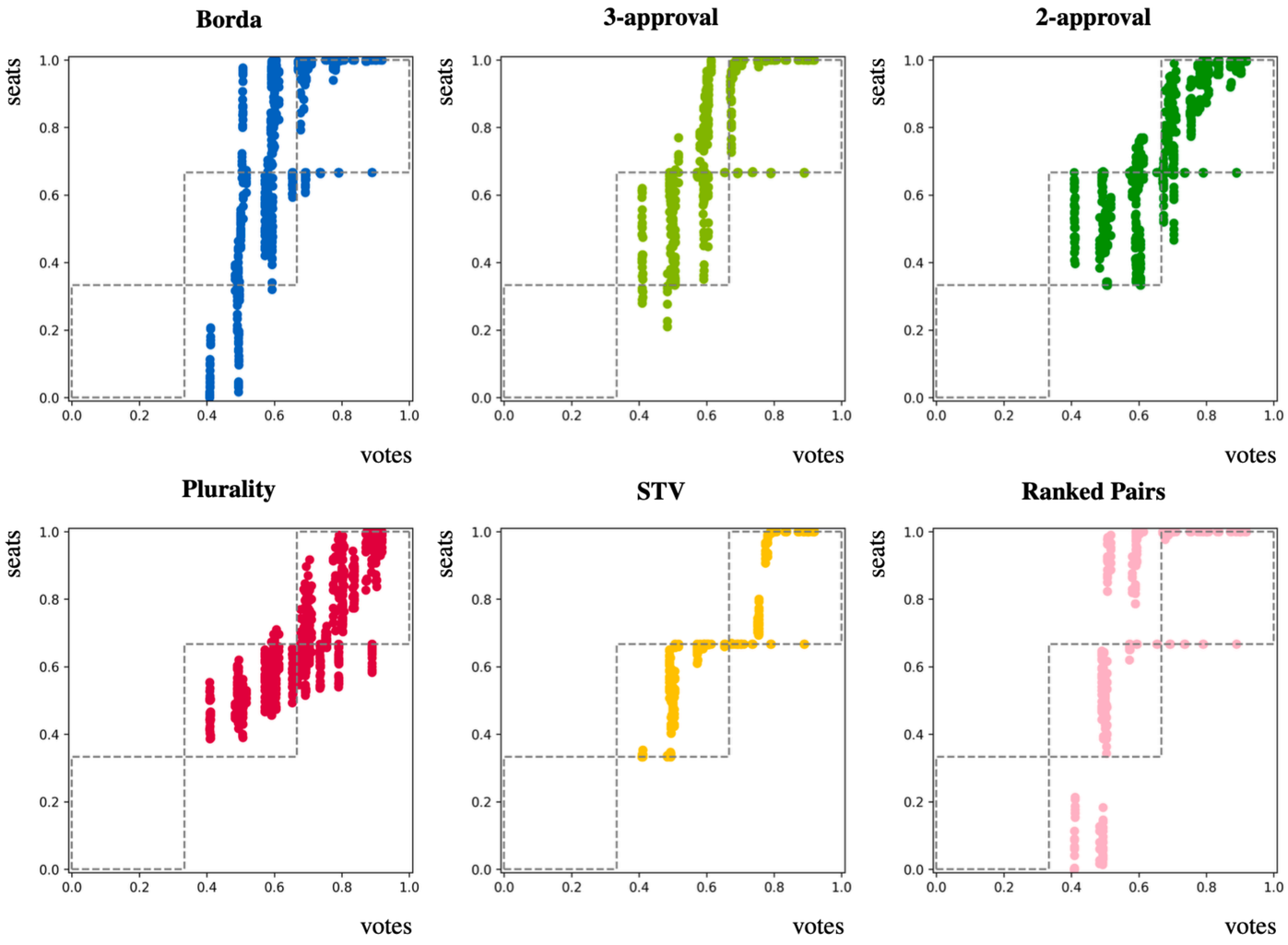
# Scottish empirics

Top-Right = “good” for our metrics



# So what about **fairness**?

In the box = proportional ("good")



Thank you!